ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · A6 · FLOWERING PLANTS

Parts of a flower

★☆☆
Grade 5

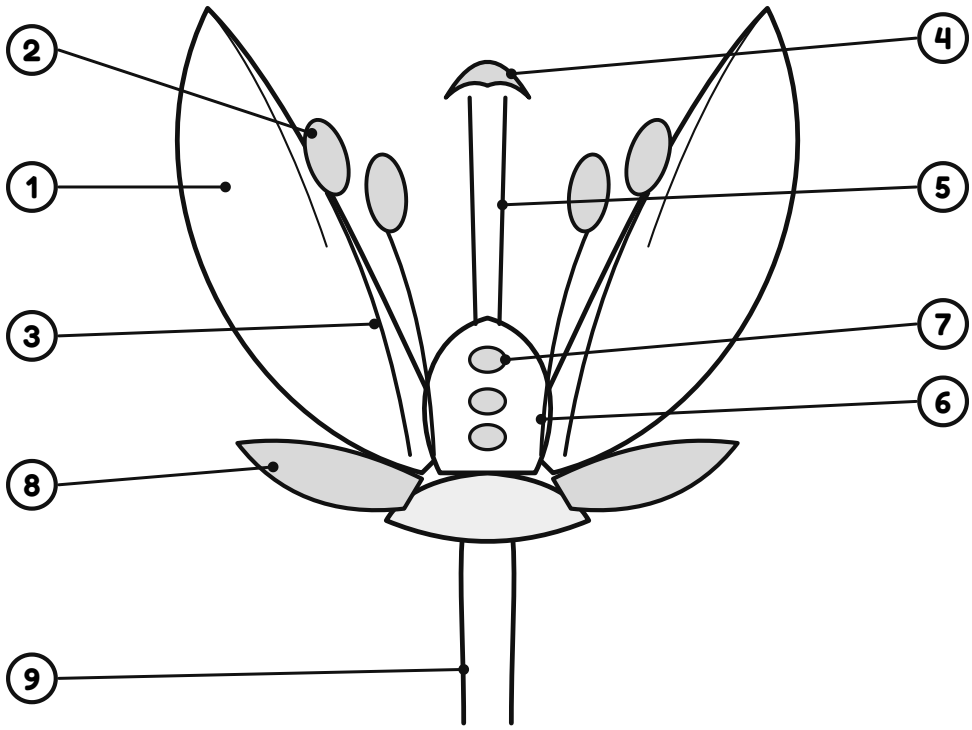
Learning goals

- Label the parts of a flower.
- Sort the parts into male and female.
- Use the key words correctly.

1 **Label the parts of this flower. Use the words in the box.** [9]

LABELS

stigma **ovule** **style** **petal** **flower stalk** **anther** **filament** **ovary** **sepal**



1. petal
2. anther
3. filament
4. stigma
5. style
6. ovary
7. ovule
8. sepal
9. flower stalk

2 Sort the flower parts into the table. [4]

SORT THESE

petal **style** **filament** **anther** **sepal** **stigma** **ovule** **ovary**

Male parts	Female parts	Other parts
anther	stigma	petal
filament	style	sepal
	ovary	
	ovule	

3 Match each word to its meaning. [6]

Draw a line to match each pair, or write the letter.

pollination ●	→ F	● **A** seeds are carried away from the parent plant
fertilisation ●	→ E	● **B** a seed starts to grow
seed dispersal ●	→ A	● **C** a powder made by the anther
germination ●	→ B	● **D** a sweet liquid that attracts insects
nectar ●	→ D	● **E** a male cell joins with a female cell in an ovule
pollen ●	→ C	● **F** pollen moves from an anther to a stigma

4 True or false? Tick the box. If a sentence is false, write the correct version. [5]

Statement	True	False	If false, correct it
1. The anther is a female part of the flower.		✔	The anther is a male part.
2. Pollen contains male cells.	✔		
3. After fertilisation, the ovary grows into a fruit.	✔		
4. Wind-pollinated flowers usually have big, bright petals.		✔	They are usually small and dull.
5. Each fertilised ovule becomes a seed.	✔		

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · A6 · FLOWERING PLANTS

From flower to new plant

★★☆
Grade 5

Learning goals

- Put the stages of a life cycle in order.
- Compare insect- and wind-pollinated flowers.
- Explain how seeds are shaped for dispersal.

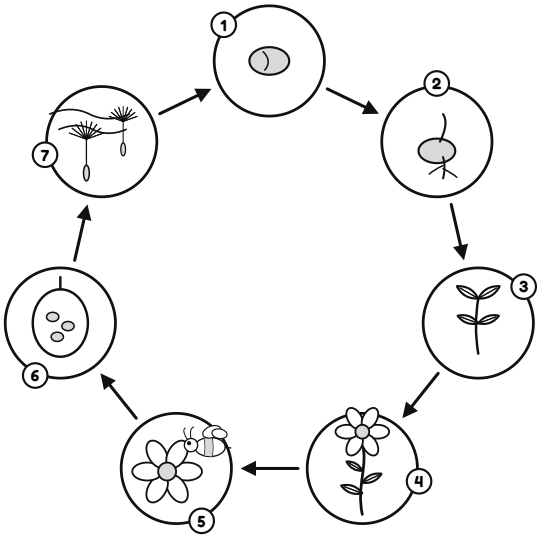
1 **These sentences describe what happens after a bee visits a flower. Number them 1 to 6 in the right order.** [3]

5	The ovule becomes a seed and the ovary becomes a fruit.
1	A bee carries pollen from another flower.
6	The fruit is eaten and the seed is dropped far away.
4	The male cell joins the female cell in the ovule.
2	Pollen lands on the sticky stigma.
3	A pollen tube grows down the style to the ovary.

2 **Label the stages of the life cycle.** [7]

LABELS

fertilisation and fruit **seed dispersal** **pollination** **seed** **flowering** **germination** **young plant**



1. seed
2. germination
3. young plant
4. flowering
5. pollination
6. fertilisation and fruit
7. seed dispersal

3 **Read the descriptions of flowers A and B.** [4]

Flower A	Flower B
Tiny green flowers, no smell, anthers hanging outside, feathery stigmas, lots of light pollen.	Large purple petals, sweet smell, nectar, sticky pollen.

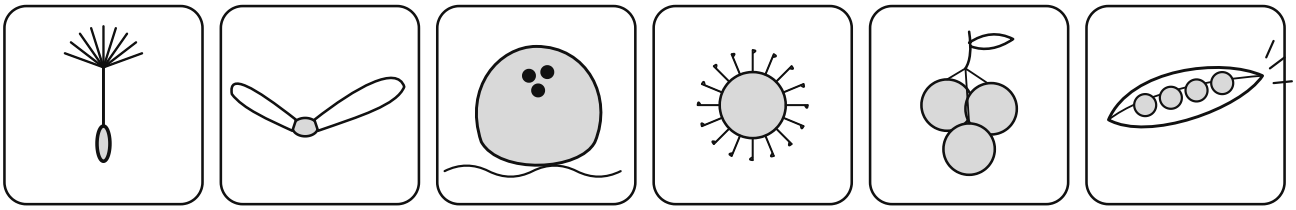
a) Which flower is pollinated by insects? Give two reasons.

Flower B: it has large bright petals, a sweet smell, nectar and sticky pollen.

b) Why does flower A make so much pollen?

Most pollen blown by the wind is lost, so it must make a lot for some to reach a stigma.

4 **Look at the seeds A to F. Complete the table.** [6]



A B C D E F

Seed	How is it dispersed?	Which feature helps?
A	wind	parachute of hairs
B	wind	wings make it spin and fall slowly
C	water	it floats (air inside)
D	animals (fur)	hooks that catch on fur
E	animals (eaten)	juicy, tasty fruit
F	explosion	pod dries and bursts open

5 **Why is it important for seeds to be dispersed away from the parent plant?** [2]

So the young plants do not compete with the parent (and each other) for light, water, space and nutrients; it also helps the plant spread to new places.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

INVESTIGATE · A6 · FLOWERING PLANTS

Spinning seeds: a paper helicopter test

★★☆ Grade 5

Learning goals

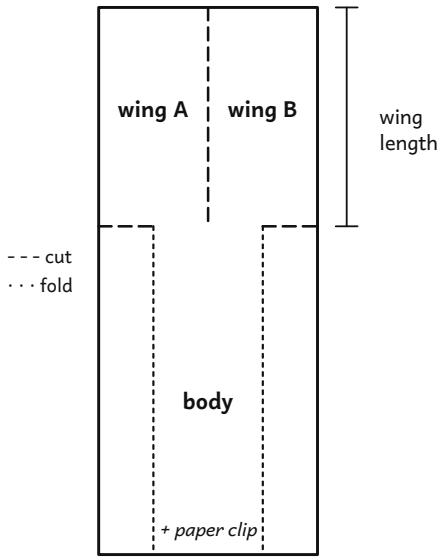
- Use a model to investigate wind dispersal.
- Identify variables in a fair test.
- Calculate averages, spot a result that does not fit and draw a conclusion.

Sycamore and other winged seeds spin as they fall. Spinning slows them down, so the wind has more time to carry them away. Class 5B made paper helicopters as **models** of winged seeds. They asked:

Does wing length change how long a paper helicopter takes to fall?

You will need

paper, scissors, ruler, paper clip, stopwatch, a safe place to drop from 2 m



Paper helicopter template

1 **Name the variables in this investigation.** [3]

Variable	In this investigation
The variable we change (independent)	the wing length
The variable we measure (dependent)	the time taken to fall (seconds)
Two variables we keep the same (controlled)	drop height, type and size of paper, body length, paper clip, how it is dropped

2 **Here are the class results. Each helicopter was dropped three times.** [5]

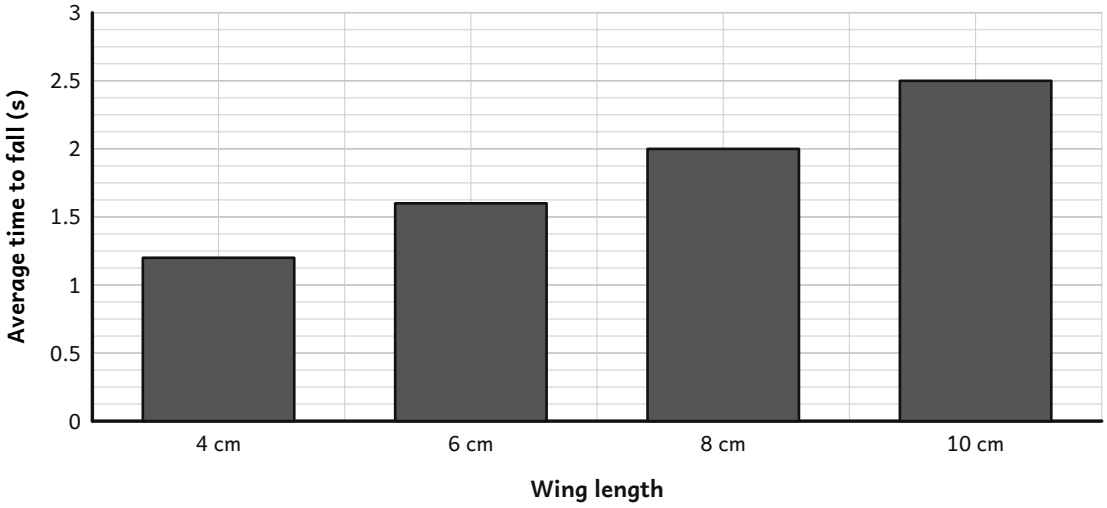
Wing length (cm)	Drop 1 (s)	Drop 2 (s)	Drop 3 (s)	Average (s)
4	1.2	1.3	1.1	1.2
6	1.6	1.5	1.7	1.6
8	2.0	2.1	1.9	2.0
10	2.4	1.5	2.6	2.5 (leaving out 1.5)

a) Work out the average for each wing length and write it in the table.

b) Which result does not fit the pattern? Suggest why it happened.

1.5 s for the 10 cm helicopter. It may have been dropped badly, hit something or the stopwatch was started late.

3 **Draw a bar chart of the average times.** [4]



4 **Write a conclusion. Use the words *longer* and *slower*.** [2]

The longer the wings, the slower the helicopter falls (the longer it takes to reach the ground).

5 **How does this help a real sycamore tree?** [2]

Seeds that fall slowly stay in the air longer, so the wind can carry them further from the parent tree.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · A6 · FLOWERING PLANTS

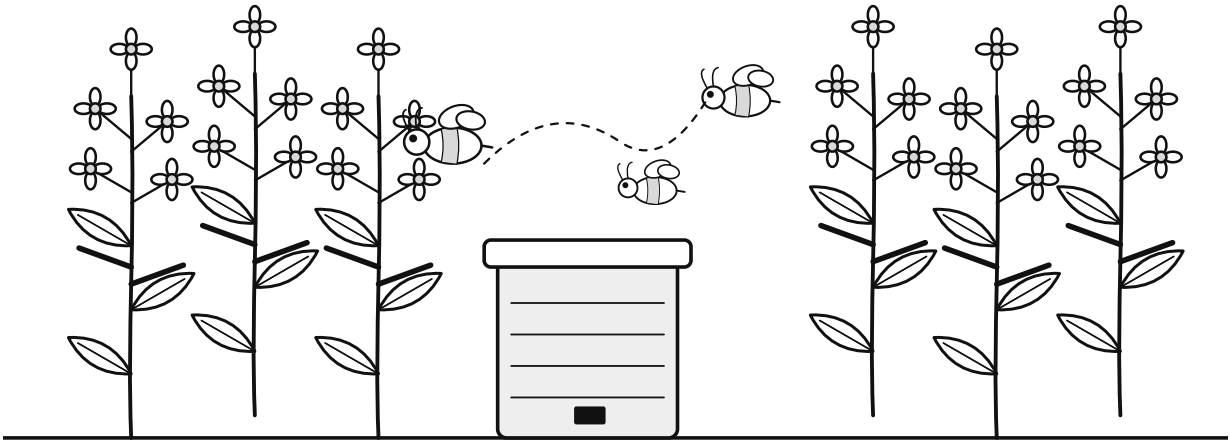
Bees, mustard and honey

★★★
Grade 5

Learning goals

- Explain how plants and animals depend on each other.
- Use data to compare results.
- Consider different points of view and suggest action.

Science | Maths | Geography | Economics | Ethics



bee box (hive)

From November to January, much of northern and central Bangladesh turns bright yellow with mustard flowers. Farmers grow mustard for its seeds, which are pressed to make **mustard oil** for cooking.

Ayesha is a beekeeper from Sirajganj. Every winter she moves 40 wooden bee boxes to the edge of the mustard fields. "My bees collect nectar from the flowers and turn it into honey," she says. "The farmers like it too, because their mustard plants make more seeds when bees visit."

But there is a problem. Some farmers spray **pesticides** on their crops to kill pests. If the spray is used while the flowers are open and bees are flying, it can kill the bees too. Ayesha now talks to farmers so they spray in the late evening, when bees are back in their hives, or use other ways to control pests.

This story is based on real beekeeping in Bangladesh; the names and numbers are examples for learning.

1 **Explain how the bees and the mustard plants depend on each other.** [2]

The mustard flowers give the bees nectar (food) to make honey. The bees carry pollen between flowers, pollinating them so the plants make more seeds.

2 **Farmer Karim grew mustard in two fields of the same size, with the same seeds, soil and water. Field A was next to Ayesha's bee boxes. Field B was far away from any bees.** [4]

	Field A (near bees)	Field B (no bees nearby)
Mustard seed harvested (kg)	180	150

a) How many more kilograms of seed did Field A produce?

180 – 150 = 30 kg more

b) Suggest a scientific reason for the difference.

More flowers were pollinated by bees in Field A, so more ovules were fertilised and became seeds.

c) Why was it important that the fields were the same size, with the same seeds, soil and water?

To make it a fair test: the only difference was the bees, so the bees caused the difference.

3 **Different people see the problem in different ways. What might each person say about spraying pesticides?** [3]

Person	What they might think
A farmer with pests on his crop	Pests eat my crop; I need to spray to protect my harvest.
Ayesha, the beekeeper	Spraying in the day kills my bees and I lose honey and income.
A family buying mustard oil and honey	We want enough oil and honey at a fair price, and food that is safe.

4 **Suggest two actions that could help both farmers and bees.** [2]

Spray only in the evening when bees are in their hives; use fewer or safer pesticides; plant flower strips for bees; farmers and beekeepers agree on spraying times.

Take it further

Maths: if Field A grew 180 kg and oil is about one third of the seed's mass, about how much oil could be pressed? (180 ÷ 3 = 60 kg). **Geography:** find Sirajganj on a map of Bangladesh.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · A6 · FLOWERING PLANTS

Show what you know

★★★
Grade 5

Learning goals

- Use key vocabulary.
- Design a model that copies how an animal disperses seeds.
- Answer test-style questions.

1 **Find these ten words in the grid.**

Y	S	H	N	I	T	U	H	D	K	F
J	E	A	E	P	O	L	L	E	N	O
A	E	N	C	B	L	Y	S	C	J	W
S	D	T	T	K	A	K	E	J	K	E
T	V	H	A	O	N	U	P	W	F	C
I	J	E	R	U	V	G	A	O	R	O
G	J	R	P	E	I	A	L	M	U	V
M	U	F	K	E	T	A	R	L	I	U
A	B	O	F	L	T	L	J	Y	T	L
T	D	O	G	N	G	A	D	B	B	E
B	F	U	W	E	U	B	L	S	P	T

anther fruit nectar ovary ovule petal pollen seed sepal stigma

2 **Design challenge. Burrs stick to animal fur so their seeds travel. Design a model seed that will stick to a woollen sock and stay on while you walk 10 steps.** [4]

Draw and label your design

A good design: a small ball (seed) covered with hooks, e.g. bent pipe-cleaner or Velcro hooks; labels show the hooks and the seed.

a) What materials will you use?

e.g. clay ball, toothpicks bent into hooks, Velcro, pipe cleaners

b) How will you test if it works?

Press it onto a sock, walk 10 steps, count how many models are still attached; repeat 3 times.

3 **Which part of a flower becomes the fruit?** [1]

- ○ petal
- ● **ovary**
- ○ stigma
- ○ anther

4 **A coconut can float across the sea. Which way is it dispersed?** [1]

- ○ wind
- ● **water**
- ○ animals
- ○ explosion

5 **Rashid says, "Seeds need light to make a flower." Explain why he is wrong and what flowers are really for.** [2]

Flowers do not grow from light alone; a flower is the part of the plant used for reproduction. It makes pollen and ovules so that seeds can form after pollination and fertilisation.